Predicting Urban Taxi Demand and Smart Charging Strategies for Electric Fleets

Advanced Analytics and Applications

Team 1

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1 Introduction

1.1 Problem Description

Urban taxi demand is highly uneven across both space and time. In a city such as Chicago, trip volumes vary by neighborhood, hour of day, weekday, weather conditions, airport activity, commuting patterns, and local points of interest. For a fleet operator, this creates a practical planning problem: vehicles should be available where passengers are likely to request rides, while unnecessary idle time, empty repositioning, and inefficient charging should be reduced.

This project uses Chicago taxi trips as a proxy for urban ride-hailing demand. The core problem is to understand and predict how many trips will start in different parts of the city during a given time window. In this report, demand is therefore defined as the number of taxi pickups per spatial unit and time interval.

1.2 Business Goal

The business goal is to support better operational decisions for a taxi or ride-hailing fleet operator. More accurate knowledge of where and when demand occurs can help operators position vehicles in high-demand areas, reduce passenger waiting times, avoid unnecessary idle time, and improve the utilization of available vehicles. For an electric fleet, this planning task is connected to charging decisions: vehicles need enough energy to serve expected demand, but charging should be scheduled in a cost-efficient way.

From a managerial perspective, the project aims to turn historical trip data into practical recommendations for fleet positioning, demand planning, and charging strategy. The final value of the analysis is not only whether a model predicts demand accurately, but whether the results can inform decisions that improve service reliability and operational efficiency.

1.3 Data Mining Goal

The data mining goal is to build a spatio-temporal demand prediction task from raw taxi trip records and external contextual data. Individual trips are aggregated into panels by time window and spatial unit, and the target variable is the number of pickups in each panel cell. Calendar variables, weather data, and spatial identifiers are used as explanatory features.

The project evaluates whether machine learning models can predict aggregate taxi demand at useful resolutions. In particular, Support Vector Machine models and neural networks are trained and compared against simple time-profile baselines. Model performance is assessed using appropriate error metrics and benchmarking, with attention to whether the models perform well not only overall but also in the high-demand areas that matter most for fleet operations. In a second analytical task, a reinforcement learning approach is used to study how charging decisions for an electric taxi can be optimized under operational constraints.

2 Data Description

The analysis combines taxi trip records, hourly weather observations, spatial reference data, and electricity prices. Taxi and weather data cover 2024-01-01 through 2026-05-31. The period ends in May because the city portal reports recent trips with a delay; including the incomplete June data would create an artificial decline in demand.

2.1 Taxi Trip Data

The main source is the Chicago Taxi Trips dataset from the City of Chicago Data Portal (dataset `Taxi Trips (2024-)`). Each row represents one reported trip by a licensed Chicago taxi. Before cleaning, the extract contains 16,086,196 rows and 21 columns. It includes trip timestamps and duration, distance, fare components, payment type, taxi company, anonymized identifiers, and pickup and dropoff locations.

Locations are reported through the identifiers and centroids of Chicago's 77 community areas and 878 census tracts. Census tracts provide finer spatial detail but are frequently masked in low-volume areas for privacy. Where tract information is unavailable, the coordinate columns may instead contain community-area centroids. This mixed spatial resolution limits fine-grained interpretation and is considered during data preparation and spatial analysis.

2.2 Weather Data

Hourly weather observations are obtained from the Open-Meteo archive API. The dataset contains 21,168 observations and nine variables for the same period as the taxi data. Variables include temperature, apparent temperature, precipitation, snowfall, wind speed, cloud cover, and WMO weather conditions. Weather is joined to the demand data by timestamp and provides exogenous features that can also be known for future prediction periods.

2.3 Spatial Reference Data

Official boundary files provide polygons for Chicago's 77 community areas, 878 census tracts, and the city boundary. They support mapping, spatial joins, and demand aggregation at different geographic resolutions. Point-of-interest data from OpenStreetMap add locations such as transport facilities, amenities, and tourism-related places, helping to characterize areas that may generate taxi demand.

2.4 Electricity Price Data

The reinforcement-learning analysis uses EPEX SPOT day-ahead prices obtained through the Fraunhofer ISE Energy-Charts API (Fraunhofer Institute for Solar Energy Systems ISE, 2026). The data cover the German–Luxembourg bidding zone from January through March 2025 and are converted from EUR/MWh to EUR/kWh. For each of the 90 days, prices between 14:00 and 16:00 are transformed into eight 15-minute values matching the charging decisions. These published prices are observable to the price-aware charging agent; the price ultimately billed is simulated by adding a random short-term deviation. Details of this transformation and the simulated price process are given in the reinforcement-learning section.

2.5 Granularity and Limitations

The cleaned taxi data contain 14,465,662 rows and 27 columns. For spatial analytics and predictions, taxi trip observations are aggregated from individual trips into spatio-temporal demand panels, using community areas or census tracts and hourly or four-hour windows. Key limitations are privacy-related location masking, the reporting lag in recent portal data, mixed spatial precision, and the use of licensed taxi trips as a proxy for wider ride-hailing demand.

3 Data Preparation

The raw taxi export contains 16,086,196 rows across 21 columns. The cleaning pipeline concatenates the 2024, 2025, and 2026 CSV exports, applies the steps below, and writes the result out as a single Parquet file. In total, 10.1% of trips (1,620,534 rows) are removed:

1. **Duplicate removal.** Trips are deduplicated on `trip_id` before any other filtering (775 rows removed), keeping the first occurrence. Rows without a `trip_id` are set aside first and rejoined afterward, since `drop_duplicates` would otherwise treat several genuine but ID-less trips as duplicates of each other.

2. **Handling missing spatial identifiers.** Chicago reports location at two resolutions: `census_tract` (fine-grained) and `community_area` (77 coarse zones covering the whole city). Tracts are missing for over half of trips — not randomly, but because the city masks any tract with too few trips, for privacy. Community areas are far more complete (98.2% pickup, 91.6% dropoff), since they're only missing when a trip starts or ends outside city limits. Rather than dropping those out-of-city trips, we keep every row and add `pickup_flag`/`dropoff_flag` columns marking whether an area is present, so later spatial analyses can filter on the flags without losing the trips for everything else.

3. **Resolution-consistent centroids.** The centroid latitude/longitude columns silently mix resolutions: where a tract exists, they hold its centroid; where it doesn't, they fall back to the community area's centroid. We derive a clean area-to-centroid mapping from exactly the rows where the raw coordinate already is the area centroid and join it back as separate `*_ca_centroid_lat/lon` columns, giving a resolution-consistent coordinate for any area-level aggregation.

4. **Removal of incomplete trip records.** Trips missing `fare`, `trip_total`, `trip_seconds`, or `trip_miles` are dropped (34,687 rows), since no meaningful analysis is possible without them.

5. **Outlier and plausibility filtering.** Negative `tips`/`tolls`/`extras` would be removed, though none occur in practice — zero is legitimate, for example for cash trips that report no tip. Trips below Chicago's legal minimum fare ($3.25 flag-drop) or shorter than 60 seconds are treated as test or aborted entries and removed, together with trips above the 99.9th percentile of distance (43.0 miles) or duration (165 minutes). These bounds account for the bulk of all removed rows (1,585,072 of 1,620,534).

`payment_type` and `company` are left untouched throughout; their missing values remain as `NA` and are handled per analysis.

The cleaned dataset has 14,465,662 rows and 27 columns, spanning 2024-01-01 to 2026-05-31. Table 1 in the appendix gives the full column-by-column reference: dtype, completeness, distinct-value count, and description for every column in the cleaned file.

4 Descriptive (Spatial) Analytics

This section analyzes 14,465,662 cleaned taxi trips from January 2024 to May 2026. Demand is defined as observed pickups and therefore represents served taxi demand rather than all latent ride-hailing demand.

4.1 Temporal and Trip Patterns

Demand follows a clear daily and weekly cycle. The lowest-volume hour is 03:00, with an average of 62.96 pickups per observed day, while demand peaks at 17:00 with 1,194.63. Thursday is the busiest day (18,856 trips per day) and Sunday the quietest (12,591; Table 2). Commuting, business activity, shopping, and after-work travel are plausible explanations for the weekday and late-afternoon concentration. A separate weekend profile does not show a sufficiently large overnight peak to offset the generally lower weekend volume.

Temporal resolution determines how much of this pattern remains visible. Hourly bins retain the exact 17:00 peak; two- and three-hour bins progressively smooth local fluctuations; four-hour bins provide a more stable planning signal, with the highest volume from 16:00–20:00; and daily aggregation removes the intraday cycle. Hourly data are therefore preferable for short-term vehicle positioning, whereas four-hour windows are more suitable for fleet allocation.

Trip distance, duration, and price are strongly right-skewed (Table 3). A typical trip has a median distance of 3.85 miles, lasts 16 minutes, and has a metered fare of USD 15.25. The corresponding means are higher because of a smaller number of long airport and cross-city trips. Trip composition also changes over the day: the lowest mean fare and shortest median distance occur at 08:00 (USD 18.54 and 2.26 miles), whereas the highest mean fare occurs at 04:00 (USD 32.62) and the longest median distance at 05:00 (11.67 miles). A larger share of airport-oriented trips is a plausible explanation, although the descriptive analysis does not separate this effect from surcharges or traffic conditions.

Idle time is not directly recorded. It is approximated as the gap between a taxi's trip end and its next recorded trip start. Negative gaps are excluded, and gaps above four hours are treated as probable off-shift periods. Under this rule, 12,072,245 transitions yield a mean idle-time proxy of 44.49 minutes and a median of 30 minutes. Because the estimate changes with the cutoff, alternative thresholds are reported in Table 4 and visualized in Figure 1. It should be interpreted as a utilization proxy, not a direct measurement of vehicle availability.

4.2 Spatial Patterns and Resolution

Pickup and drop-off demand are concentrated in central Chicago and at the airports (Table 5). Near North Side leads both pickups (3,264,625) and drop-offs (3,528,594). O'Hare is strongly origin-oriented, with 2,985,521 pickups but only 579,319 drop-offs. Together, Near North Side, O'Hare, the Loop, and Near West Side account for 71.06% of pickups with a reported Community Area. Downtown employment, hotels, tourism, nightlife, rail connections, and airport flight schedules are plausible demand generators.

The leading zones vary over time (Table 6). Near North Side dominates pickups from 04:00–20:00, while O'Hare leads from 00:00–04:00 and 20:00–24:00. Near North Side leads most drop-off blocks, with the Loop leading from 04:00–08:00. Origins and destinations must therefore be modeled separately, and fleet positioning should change over the day.

The three spatial resolutions provide complementary views. Community Areas are complete, recognizable operating zones but hide local variation. Census Tracts offer finer detail, although the existing analysis retains only the 6,245,422 trips (43.17%) for which both tract identifiers are available. H3 resolution 7 provides a uniform grid, but it cannot recover street-level precision because the anonymized data contain only 573 distinct pickup and 687 drop-off coordinate pairs. Only one H3 size is tested; resolution variation is primarily examined through Community Areas, Census Tracts, H3, and the one-, two-, three-, four-hour, and daily temporal bins. Finer resolutions reveal local peaks but are more sparse and privacy-sensitive.

4.3 Gaussian-Mixture Hot Spots

A Gaussian Mixture Model (GMM) complements the discrete maps with a parametric spatial density estimate. It is fitted to 2,787,934 tract-resolved pickups from 2025 at 421 locations inside Chicago. Each component provides a center, spatial extent, and share of pickup demand. A 100-meter covariance floor prevents components from collapsing onto anonymized centroids.

The BIC curve has an elbow at six components, but the final model uses three operationally interpretable and spatially separated zones: the extended city center (76.3% of tract-resolved pickups), O'Hare (21.0%), and Midway/Garfield Ridge (2.6%). Additional components mainly divide the contiguous downtown area near the approximately 600-meter resolution limit. The fitted density and component estimates are provided in Figure 2 and Table 7.

The hot-spot locations remain stable over the day, but their importance changes. The city-center share is approximately 80% during 07:00–19:00 and 61% at night; O'Hare rises from 18% to 36%. This supports rebalancing between a stable set of downtown and airport zones rather than uniform city-wide coverage.

Overall, the notebooks answer the descriptive task across time, trip characteristics, prices, idle-time proxies, origins, destinations, spatial resolutions, and GMM hot spots. The principal limitations are centroid anonymization, incomplete Census Tract coverage, one fixed H3 size, and the threshold-dependent idle-time definition.

5 Predictive Analytics

5.1 Problem Description

We predict taxi pickup demand – the number of trips originating in a spatial unit within a fixed time window. Two model families are used: Support Vector Regression (SVR) and feedforward neural networks (NN). Both are evaluated at community-area resolution in 4-hour and 1-hour windows; census-tract resolution (4-hour) is evaluated with the neural network only. Community area at 4-hour resolution is treated as the default setting throughout, and the other resolutions are presented as deviations from it.

5.2 Modeling Approach

Both families share the same task framing: a single **pooled model per experiment**, trained across all spatial units at once, with spatial identity encoded as one-hot fixed-effect dummies rather than fit as separate per-unit models. Two families were tried for complementary reasons: SVR (linear and kernelised) is cheap to fit and, in its linear form, interpretable through its coefficients; a feedforward network can in principle learn non-linear interactions between spatial identity and time features that a linear model cannot without hand-built interaction terms.

The central obstacle is that demand is not one distribution but dozens. Per-unit demand varies roughly three orders of magnitude across community areas, and each unit has its own daily and weekly rhythm: an airport runs around the clock on flight schedules, while a downtown office district peaks at commute hours and goes quiet overnight. A pooled model must represent all of that with one set of parameters. This is hardest for a linear SVR, whose single global coefficient vector can only find one compromise direction across every unit at once; a kernel SVR could in principle represent the heterogeneity, but its training cost scales quadratically to cubically in the number of rows, which forces the kernel variants onto a 10,000-row subsample.

Both use the same exogenous feature set: weather (temperature, precipitation, snowfall, wind, cloud cover), cyclical time encodings (sine/cosine pairs for hour-of-day and weekday, two harmonics each), month, and weekend/holiday indicators. Per the constraints of this course, no lagged or autoregressive demand features are used. Both regress on `log1p(count)`, with predictions back-transformed via a clipped `expm1`, since raw demand is heavily right-skewed (Figure 3).

5.3 Validation Strategy

Every model is judged against a **profile baseline**: the train-set mean demand for each (hour-of-day, weekday, month) bucket, computed separately for each spatial unit. Because it is fitted per unit, the baseline sidesteps the heterogeneity problem described above – it never has to compromise one parameter set across areas whose demand differs by three orders of magnitude – which is what makes it a demanding reference throughout this section. Both model families use the same chronological holdout: training on 2024-01-01 through 2025-12-31 and testing on 2026-01-01 through 2026-05-31, over a complete zero-filled grid of every unit crossed with every time window, so that evaluation includes genuine zero-demand periods and not only observed trips.

The baseline already explains a large share of the variation in the data. At the default community-area, 4-hour resolution its pooled test R^2 is 0.933 (Table 8), and it tracks the daily and weekly demand cycle closely across an evaluation week (Figure 4); the same holds at the finer settings (pooled R^2 0.918 at 1-hour and 0.896 at census-tract level). Per-unit fit is far more variable – median per-area R^2 is only 0.208 at 4-hour resolution, with 21 of 77 areas negative (Figure 5) – because the pooled figure is dominated by the few highest-volume units.

Two headline metrics summarise each model; both are defined formally in Section 8. **Demand-weighted skill** is a per-unit ratio of the model's MAE to the baseline's MAE (Equation 1), aggregated with weights proportional to each unit's median demand (Equation 2) so that 0 = tied with the baseline, positive = better, negative = worse; it cancels each unit's own scale and is comparable across resolutions. **Weighted RMSE** (Equation 3) measures error magnitude in trips per window, with high-volume rows up-weighted, and is the metric used to compare the two model families directly.

5.4 Results

Community area at 4-hour resolution is the default setting; the other settings follow as deviations from it.

5.4.1 Support Vector Regression – Community Area, 4-Hour

Ten SVR configurations were run at this resolution (Table 9). A pooled `LinearSVR` reaches demand-weighted skill −0.827 (weighted RMSE 254.94); the sample-weighted and demand-tiered variants stay strongly negative. Kernel SVR tuned on a 10,000-row subsample scores higher (tuned RBF −0.232, polynomial −0.221) but remains below the baseline. Fully separate per-area models, fit only for the 20 areas above a median-demand threshold of 10 trips/window, score higher still: the tuned per-area RBF SVR is the single configuration with positive demand-weighted skill, +0.012 (weighted RMSE 143.80). Figure 6 shows the pooled variants over the March week in Community Area 8, and Figure 7 maps pooled `LinearSVR` skill across the 77 areas. All ten configurations are collected in Table 9.

5.4.2 Neural Network – Community Area, 4-Hour

Two networks were run: an untuned shallow network (one hidden layer) and a deeper network selected by a one-factor-at-a-time architecture search. Their test scores are in Table 10. The shallow network reaches demand-weighted skill −0.028 (weighted RMSE 144.96) and the tuned deep network −0.027 (weighted RMSE 145.93), against the baseline's 139.02; both sit just below the baseline on skill. Figure 8 shows the three models tracing demand across the March evaluation week in the highest-demand area (Community Area 8). The per-area skill map (Figure 9) is at or above zero across most of the city, with O'Hare (community area 76) the one clearly negative area (skill −0.227).

5.4.3 Spatial and Temporal Resolution Variations

Community area, 1-hour. At 1-hour resolution the same 77 areas are split across 24 hourly buckets instead of 6, quadrupling the rows and the baseline's granularity. Test scores for both families are in Table 11. For SVR, demand-weighted skill of the pooled `LinearSVR` falls to −1.016 and the best tuned kernel (RBF) reaches −0.331. For the neural network, the tuned deep configuration selected at 4-hour resolution (reused unchanged) reaches demand-weighted skill +0.062 (weighted RMSE 36.26, against the baseline's 39.46) – the only positive NN result in the study – while the shallow network stays negative (−0.048). March-week traces appear in Figure 10 and Figure 11, and per-area skill maps in Figure 12 and Figure 13; scores for both families are in Table 11.

Census tract, 4-hour. Only 44.6% of trips carry a true tract-level location; Chicago masks the rest to the community-area centroid whenever a tract had too few trips in a period, for privacy. To recover realistic tract-level demand, tract-precision coverage was first measured per community area, and only areas retaining at least 5% of their trips at tract precision were kept – a natural gap in the data separates areas at 4.85% from those at 7.17% – leaving **179 of 878 tracts across 11 community areas** in scope (Figure 14); the remaining areas fall back to the community-area model. Within scope, per-window counts were built from tract-precision trips only and then scaled up by each area's inverse retention rate, so the observed fraction is extrapolated to an estimate of each tract's total demand. This keeps the masked points from spiking demand artificially onto whichever tract holds a community area's centroid.

Only neural networks were run at this resolution (Table 12). The shallow network reaches demand-weighted skill −0.098 (weighted RMSE 113.77, against the baseline's 100.99) and the deep network −0.340. March-week traces for the top tract are in Figure 16 and the per-tract skill map in Figure 17. SVR was not run at tract level because a per-tract one-hot encoding (up to 878 dummy variables) is computationally infeasible for kernel SVR.

5.5 Discussion

5.5.1 Model Comparison

Across every setting the profile baseline is a strong reference, and it is hardest to beat exactly where demand is highest. Both families lose demand-weighted skill specifically in the busiest areas, because a single pooled parameter set cannot give a high-volume downtown its own scale and rush-hour shape while also serving the low-demand majority, whereas the per-unit baseline never faces that compromise. The neural network has the representational capacity to learn per-area time interactions and comes closest: near break-even at 4-hour resolution and clearly positive (+0.062) once the same architecture is reused at 1-hour resolution. Pooled SVR trails further behind at every resolution, and only abandoning pooling altogether – a dedicated per-area RBF model for the 20 busiest areas – yields a positive result (+0.012). The two families' failures also share a location: the deep NN's worst area at 4-hour resolution is O'Hare, whose round-the-clock, flight-driven demand differs in *shape*, not just scale, from the commute cycle the pooled model averages toward.

Resolution matters in opposite directions for the two families. For matched SVR types, skill is worse at 1-hour than at 4-hour, so the finer grid sharpens the high-demand gap; for the neural network, the 4-hour architecture transfers to 1-hour cleanly and produces its one clear win. Census-tract results are the weakest overall: even with the inverse-retention extrapolation, the per-unit signal is thin once 179 units compete for demand, and both networks fall below the tract baseline.

5.5.2 Practical Implications

The consistent pattern – pooled models fail in the high-demand areas that matter most for allocation – points toward a targeted deployment rather than one universal model. A fleet operator does not need accurate predictions for every low-demand area; those are served about as well by the cheap profile baseline as by anything tested here. The value concentrates in the small set of highest-demand areas, and the only configurations that beat the baseline there – the per-area RBF SVR and the 1-hour deep NN – are precisely the ones that either abandon pooling or exploit a finer temporal grid. The practical recommendation is therefore to fit dedicated models for the handful of high-value areas and leave the long tail of low-demand areas to the profile baseline, which already handles them about as well as any model tried here.

6 Reinforcement Learning

6.1 Problem Setup

The task considers a taxi driver who charges her vehicle at home during a fixed afternoon window; a flat charging rate is replaced by an agent that selects the charging power for each interval, subject to the vehicle's battery capacity. The battery must cover a shift's demand, which is stochastic and unknown until departure, so recharging cost must be weighed against a heavy penalty for arriving under-charged. The resulting environment assumptions are summarized in Table 13.

6.2 Markov Decision Process

The charging problem is modeled as a finite-horizon Markov Decision Process. Without day-ahead price information, the state is

$$s_t = [SOC_t, t, C],$$

where SOC_t is the current energy level, $t \in \{0, \dots, 8\}$ is the time step, and C is battery capacity. The realized demand is deliberately excluded from the state. When day-ahead price information is included, the state is extended with the current day-ahead price and the mean remaining day-ahead price:

$$s_t^{EPEX} = [SOC_t, t, C, \pi_t^{DA}, \bar{\pi}_{\geq t}^{DA}].$$

Four discrete charging powers of 0, 5, 11, and 22 kW are used initially. Because this sparse spacing leads to degenerate behavior under the exponential cost function, a refined grid of 0, 4, 8, 12, 16, 20, and 22 kW is used in the final iteration. For a 15-minute interval,the transition is

$$SOC_{t+1} = \min(C, SOC_t + p_t \cdot 0.25).$$

The step reward is the negative charging cost. After the eighth decision, a terminal penalty is added if $SOC_8 < d$:

$$r_8 = -\text{cost}_8 - 100 \max(0, d - SOC_8).$$

A cost-reliability trade-off results: additional energy is inexpensive relative to a shortage, while the tail risk must be learned from sampled experience.

6.3 Cost Environments and Electricity Prices

Two `gymnasium` environments are used, sharing the same battery dynamics and hidden-demand process but differing in charging-cost model. In the stylized environment, the cost of selecting power p in interval t is defined as

$$\text{cost}(t, p) = \alpha_t \exp(p),$$

where α_t increases from 2.0×10^{-10} to 7.2×10^{-10} over the charging window. The small coefficients are required because exp(22) is approximately 3.6×10^9. Even charging at 22 kW in all eight intervals costs only 11.90 reward units, compared with the shortage penalty of 100 € per missing kWh, ensuring charging is always better than not charging. Consequently, the original costs of 0, 5, and 11 kW are almost indistinguishable, while 22 kW constitutes a separate, much more expensive tier. Jumps between adjacent actions are reduced by the refined grid, allowing more graduated control, although the cost function remains nonlinear. The stylized model should therefore be interpreted as an artificial power-dependent reward rather than a realistic electricity tariff.

The EPEX environment uses the day-ahead price data described in Section 2, published one day before delivery and linearly interpolated within each hour to the eight quarter-hour charging decisions. The resulting series range from -0.026 € to 0.371 € per kWh, with a median of 0.110 € per kWh. Charging is billed at the realized price

$$\pi_t = \pi_t^{DA} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, 0.015^2) \text{ €/kWh},$$

which is not observable before the decision. ε_t represents the gap between the day-ahead forecast and the delivery price and is simulated rather than drawn from real data.

6.4 Learning Algorithm and Benchmarks

Two Deep Q-Networks (DQN) are trained: a price-unaware agent in the stylized environment and an EPEX DQN in the EPEX environment. A multilayer perceptron with two hidden layers of 128 units and identical hyperparameters is used for both. Training lasts 200,000 time steps, equivalent to approximately 25,000 simulated charging days. Performance is evaluated on 1,000 paired episodes, using identical seeds so that capacity, initial SOC, demand, price day, and simulated price deviations match across strategies.

The operational benchmark is a price-blind greedy heuristic. Because demand is unknown, charging is not targeted at the realized requirement; instead, a fixed target is used:

$$\tau = \min(C, \mu + 2\sigma) = \min(C, 40 \text{ kWh}),$$

corresponding to an approximate 97.7% service level under the assumed normal distribution. At each step, the energy still missing from this target is spread over the remaining time, and the smallest sufficient charging power is selected.

An exact brute-force benchmark is used, evaluating every action sequence. For the original grid this requires $4^8 = 65{,}536$ sequences, evaluated on 100 episodes. The refined grid increases the search to $7^8 = 5{,}764{,}801$ sequences, so the EPEX oracle evaluation is restricted to 40 episodes. The realized demand and, in the EPEX setting, the realized billed prices, are known to this oracle benchmark in advance, making it a theoretical upper bound rather than an implementable policy.

6.5 Results

An episode succeeds when the final state of charge covers the realized demand ($SOC_8 \geq d$), so that no shortage penalty is added to r_8; success rate is therefore the primary reliability metric, alongside mean reward and, under EPEX prices, mean charging cost.

On the original four-action grid, the compressed cost tiers described above leave the price-unaware DQN with little reason to modulate power: it collapses onto a single action instead of spreading decisions the way greedy does (Figure 18), and it trails greedy on both success and reward. The refined seven-action grid restores graduated control and closes most of this gap (Table 14).

Under EPEX prices, with the refined grid, greedy, the price-unaware DQN, and the EPEX DQN are evaluated on the same paired episodes and billed at the same realized prices, so success rate, cost, and reward are directly comparable across policies in Table 15 (visualized in Figure 19); only the training objective differs between them. The EPEX DQN reaches the lowest cost by a wide margin, but its success rate is far below the other policies, so part of its saving reflects under-buying reserve rather than better scheduling. The price-unaware DQN nearly matches greedy's reliability but stays 18.8% more expensive on episodes where both succeed, because it was trained to minimize the stylized exponential cost rather than real per-kWh prices.

Restricted to the 752 episodes where both greedy and the EPEX DQN succeed, the EPEX DQN saves €0.63 per day (41.7%) on average, cheaper in the large majority of them and increasingly so as the day-ahead price spread within the window grows (Figure 20) – though this excludes the 24.8% of episodes it fails outright. The learned policies follow the intended qualitative pattern (Figure 21): power tracks state of charge and remaining time for the price-unaware DQN, while the EPEX DQN additionally defers to cheap intervals except when low state of charge makes charging urgent.

6.6 Operational Implications and Limitations

The price-unaware DQN's edge over greedy is an artifact of the artificial exponential reward: it evaporates under real EPEX billing, where a policy trained on the stylized cost turns out more expensive rather than cheaper (Table 15). Only the EPEX DQN, trained directly on real prices, delivers a genuine advantage.

That advantage is not free: at 75.2% success against greedy's 98.9%, part of the EPEX DQN's saving is forgone safety margin rather than pure price timing, and a policy that fails to fully charge one in four vehicles is not operationally viable on its own. Before deployment it would need an explicit safety layer – for example a hard minimum-SOC floor in the final intervals – to close this reliability gap without giving up much of the timing-based saving. That saving grows with day-ahead volatility (Figure 20), so RL-based charging is worthwhile only alongside a genuinely dynamic electricity tariff.

7 Conclusion

The analysis indicates that a potential electric-taxi operator should begin with a conservative, zone-based planning system rather than rely immediately on complex prediction and control models. Observed taxi demand is strongly concentrated: more than 70% of spatially assigned pickups originate in Near North Side, O'Hare, the Loop, and Near West Side, and demand peaks on weekdays in the late afternoon. The Gaussian Mixture Model confirms three broader operational hot spots—the extended city center, O'Hare, and Midway—whose relative importance changes over the day. These patterns support staging vehicles near central business and transport zones and rebalancing them toward airports during evening and overnight periods.

For forecasting, complexity did not consistently improve operational performance. The historical profile baseline remained difficult to outperform, and most pooled SVR and neural-network variants achieved negative demand-weighted skill. Only selected configurations, including the deep one-hour neural network and dedicated RBF models for high-demand areas, produced small improvements. A practical architecture should therefore retain the transparent profile baseline for most areas and reserve specialized models for a limited number of high-volume zones where forecast improvements have the greatest allocation value.

The charging experiment leads to a similar recommendation. Under EPEX pricing, the price-blind greedy policy achieved a 98.9% success rate and outperformed both DQN agents in mean reward after shortage penalties were included. Although the price-aware DQN reduced electricity costs, its 75.2% success rate is insufficient for fleet operations. Price-aware optimization should therefore operate only within hard minimum-SOC and departure-energy constraints. For vehicles returning regularly to a depot, private charging offers the greatest control over availability and dynamic tariffs; public fast charging remains useful as backup. The present data do not determine the optimal number or location of chargers because depot locations, grid limits, charger occupancy, and infrastructure costs are unavailable.

Overall, the project provides a proof of concept rather than a deployment-ready system. Its main strengths are the large trip dataset, explicit spatial and temporal resolution analysis, operational baselines, and the connection between demand, forecasting, and charging decisions. Important limitations are anonymized centroid locations, incomplete Census Tract coverage, a threshold-dependent idle-time proxy, inconsistent scopes across some prediction experiments, and simplified charging assumptions. The recommended next step is an interpretable system combining demand profiles, hot-spot-specific forecasts, rule-based safe charging, and continuous data collection. Further analysis of the existing trip data should examine seasonal robustness, origin–destination flows for vehicle rebalancing, and the sensitivity of idle-time and hot-spot results to alternative temporal and spatial resolutions. More advanced optimization should follow only after acquiring complete vehicle duty cycles, depot and public-charger data, traffic conditions, and infrastructure costs.

8 Appendix: Detailed Tables and Figures

Tables and figures below are referenced from the main text and grouped by chapter, in the same order as they are cited. All Predictive Analytics figures and score tables – the March-week model comparisons, the per-area skill maps, and the per-resolution score tables – are collected here and pointed to from the Results and Validation Strategy sections. The Reinforcement Learning environment assumptions, action-grid and EPEX comparisons, and policy-behavior figures are collected the same way and pointed to from the Problem Setup and Results sections.

8.1 Cleaned Dataset: Column Reference

Every column of `data/chicago_taxi_2024_2026_clean.parquet` (14,465,662 rows), with dtype, completeness, and distinct-value count as measured on that file, plus what the column means and any caveat carried over from the cleaning steps in Section 3 (e.g. census-tract masking, the two coordinate resolutions).

Table 1: Data dictionary, cleaned dataset.

Column	Dtype	Non-Null %	Distinct	Description
trip_id	str	100.0%	14,465,662	Unique trip identifier (hash). Deduplicated in cleaning step 1.
taxi_id	str	100.0%	3,354	Anonymized vehicle identifier (hash).
trip_start_timestamp	datetime64[us]	100.0%	84,660	Local Chicago date/time the trip started.
trip_end_timestamp	datetime64[us]	100.0%	84,664	Local Chicago date/time the trip ended.
trip_seconds	float32	100.0%	9,722	Trip duration in seconds (>= 60s, capped at the 99.9th percentile).
trip_miles	float32	100.0%	4,303	Trip distance in miles (> 0, capped at the 99.9th percentile).
pickup_census_tract	float32	44.6%	149	Pickup census tract (FIPS code). Missing for ~55% of trips – privacy masking of low-volume tracts.
dropoff_census_tract	float32	43.3%	175	Dropoff census tract (FIPS code). Same privacy masking as pickup.
pickup_community_area	Int64	98.2%	77	One of 77 official community areas for the pickup. Missing means the pickup was outside city limits.
dropoff_community_area	Int64	91.6%	77	Community area for the dropoff. Missing means the dropoff was outside city limits.
fare	float32	100.0%	9,078	Metered fare in USD (>= $3.25 flag-drop minimum).
tips	float32	100.0%	3,619	Tip amount in USD. 0 is valid – cash trips report no tip.
tolls	float32	100.0%	560	Toll charges in USD. 0 is valid – most trips use no toll roads.
extras	float32	100.0%	3,624	Additional surcharges in USD. 0 is valid.
trip_total	float32	100.0%	14,616	Total amount charged in USD (>= $3.25).

Column	Dtype	Non-Null %	Distinct	Description
payment_type	category	100.0%	7	Payment method (Credit Card, Cash, Mobile, Prcard, Unknown, …). Left untouched by cleaning.
company	str	100.0%	45	Taxi affiliation / operating company. Left untouched by cleaning.
pickup_centroid_latitude	float32	98.2%	571	Pickup latitude: census tract centroid where known, else community area centroid (mixed resolution).
pickup_centroid_longitude	float32	98.2%	565	Pickup longitude, same mixed-resolution rule as latitude.
dropoff_centroid_latitude	float32	92.1%	684	Dropoff latitude, same mixed-resolution rule.
dropoff_centroid_longitude	float32	92.1%	678	Dropoff longitude, same mixed-resolution rule.
pickup_flag	int64	100.0%	2	1 if pickup_community_area is present, 0 if missing (outside Chicago).
dropoff_flag	int64	100.0%	2	1 if dropoff_community_area is present, 0 if missing (outside Chicago).
pickup_ca_centroid_lat	float32	98.2%	77	Pickup community area centroid latitude – always CA-level resolution (added in step 3).
pickup_ca_centroid_lon	float32	98.2%	77	Pickup community area centroid longitude – always CA-level resolution.
dropoff_ca_centroid_lat	float32	91.6%	77	Dropoff community area centroid latitude – always CA-level resolution.
dropoff_ca_centroid_lon	float32	91.6%	77	Dropoff community area centroid longitude – always CA-level resolution.

8.2 Descriptive Analytics: Tables and Spatial Hot Spots

Table 2: Average pickup demand by weekday.

Day	Average pickups per observed day
Monday	16,382.15
Tuesday	17,681.13
Wednesday	18,358.35
Thursday	18,856.44
Friday	17,623.75
Saturday	13,313.98
Sunday	12,591.04

Table 3: Distribution of trip characteristics.

Measure	Median	Mean
Trip distance	3.85 miles	7.09 miles
Trip duration	16.00 minutes	20.97 minutes
Metered fare	USD 15.25	USD 22.06
Total charge	USD 17.75	USD 27.01

Table 4: Sensitivity of the consecutive-trip idle-time proxy.

Maximum retained gap	Transitions retained	Share of non-negative gaps	Mean proxy	Median proxy
1 hour	9,366,946	65.77%	21.20 min	15.00 min
2 hours	10,986,502	77.14%	31.88 min	15.00 min
4 hours	**12,072,245**	**84.76%**	**44.49 min**	**30.00 min**
6 hours	12,297,062	86.34%	49.09 min	30.00 min
8 hours	12,395,975	87.03%	52.08 min	30.00 min

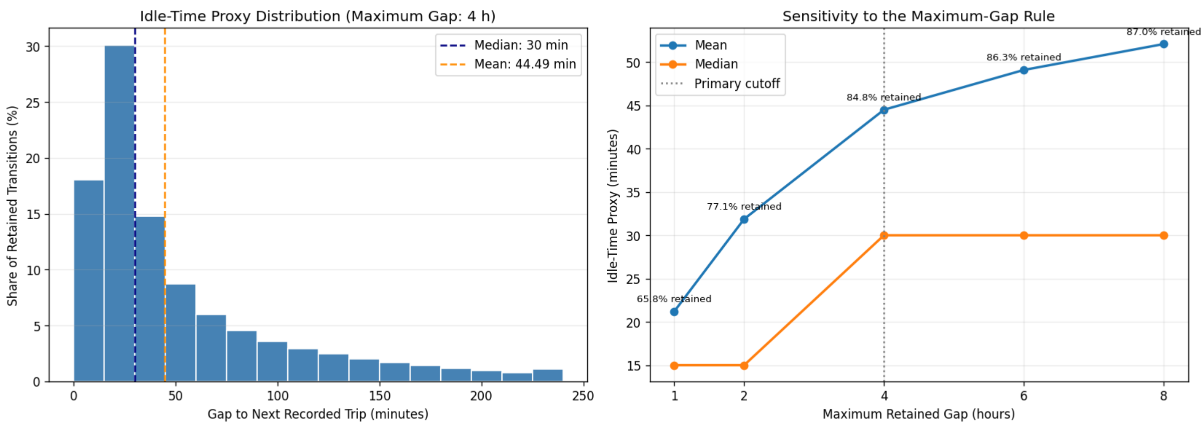

Figure 1: Idle-time proxy distribution under the four-hour rule (left) and sensitivity to the maximum retained gap (right). Percent labels show the retained share of non-negative consecutive-trip gaps.

Table 5: Highest-volume Community Areas for pickups and drop-offs.

Pickup area	Pickup trips	Drop-off area	Drop-off trips
Near North Side	3,264,625	Near North Side	3,528,594
O'Hare	2,985,521	Loop	2,452,298
Loop	2,374,677	Near West Side	1,450,354
Near West Side	1,467,354	Lake View	648,761

Table 6: Dominant Community Area by four-hour block. Pickups use start time and drop-offs use end time.

Time block	Leading pickup area	Pickup trips	Leading drop-off area	Drop-off trips
00:00–04:00	O'Hare	168,417	Near North Side	107,260
04:00–08:00	Near North Side	210,845	Loop	180,906
08:00–12:00	Near North Side	731,045	Near North Side	763,901
12:00–16:00	Near North Side	894,316	Near North Side	945,428
16:00–20:00	Near North Side	926,865	Near North Side	1,031,329
20:00–24:00	O'Hare	703,740	Near North Side	529,528

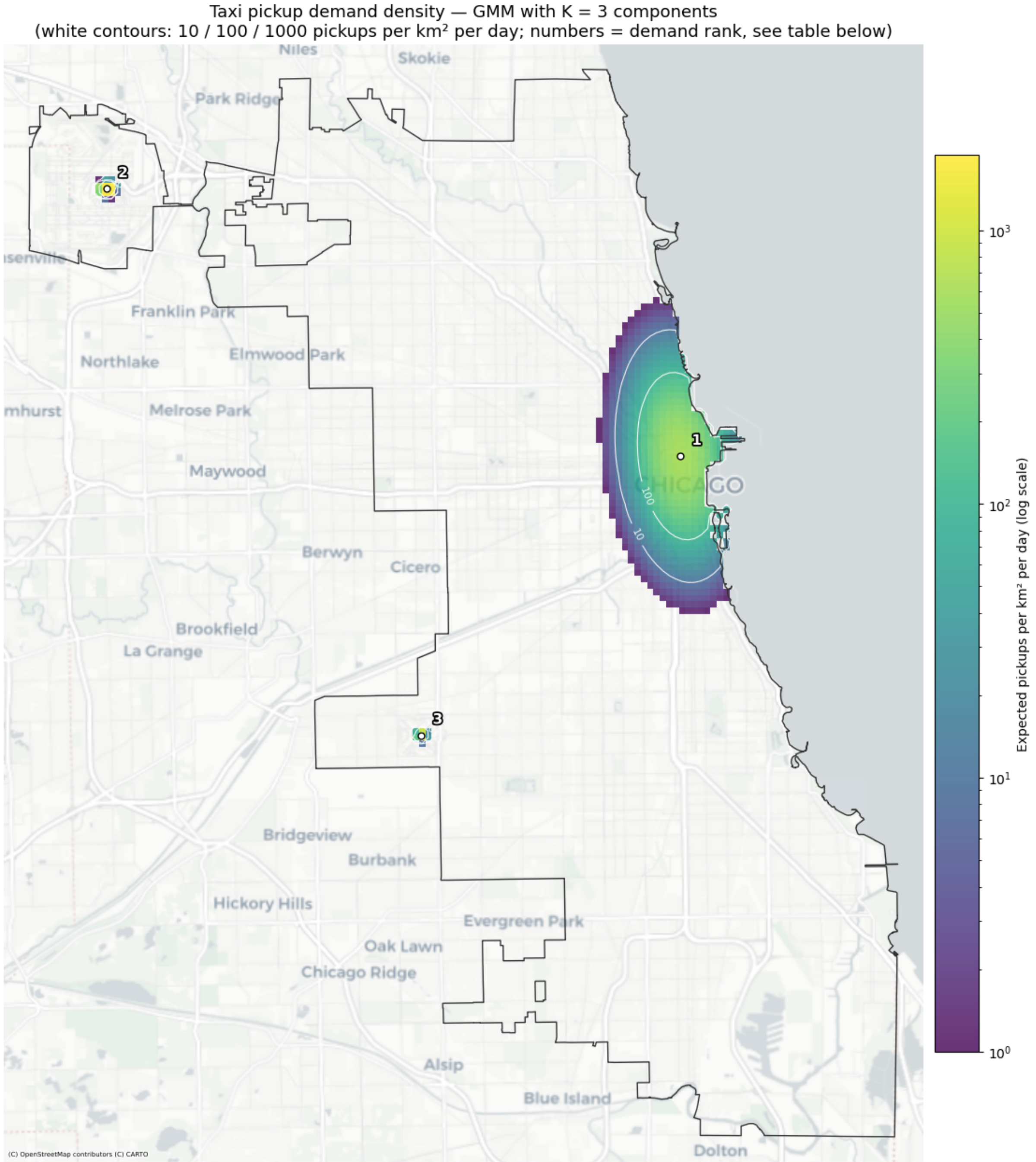

Figure 2

Table 7: Final three-component GMM hot spots.

Rank	Operational hot spot	Demand share	Tract-resolved pickups/day	Spatial extent (σ major/minor)
1	Extended city center (center in Loop)	76.3%	5,830	1,746 m / 878 m
2	O'Hare	21.0%	1,607	123 m / 100 m
3	Midway/Garfield Ridge	2.6%	201	101 m / 100 m

Rank	Operational hot spot	Demand share	Tract-resolved pickups/day	Spatial extent (σ major/minor)

8.3 Predictive Analytics: Metric Definitions

The two headline metrics used throughout the Predictive Analytics chapter are defined below. The per-unit **skill score** compares a model's mean absolute error to the profile baseline's for the same spatial unit i,

$$\text{skill}_i = 1 - \frac{\text{MAE}_{\text{model},\, i}}{\text{MAE}_{\text{baseline},\, i}} \tag{1}$$

so that 0 means tied with the baseline, 1 means zero error, and a negative value means worse than the baseline. The reported **demand-weighted skill** aggregates the per-unit skills with weights equal to each unit's median demand m_i, so the highest-volume units dominate,

$$\text{skill}_{\text{demand-weighted}} = \frac{\sum_i m_i\, \text{skill}_i}{\sum_i m_i}. \tag{2}$$

The **weighted RMSE** is a pooled, absolute-scale error over every test-set row j (a unit crossed with a time window), with y_j and $\hat{y}_j$ the actual and predicted demand and $\bar{d}_{u(j)}$ the train-period mean demand of the unit that row j belongs to,

$$\text{weighted RMSE} = \sqrt{\frac{\sum_j \bar{d}_{u(j)}\, \left(y_j - \hat{y}_j\right)^2}{\sum_j \bar{d}_{u(j)}}}. \tag{3}$$

8.4 Predictive Analytics: Baseline

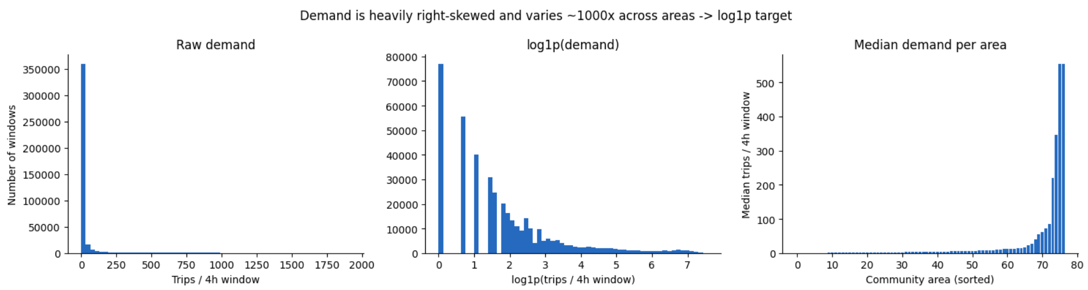

Figure 3: Raw per-window demand is heavily right-skewed, which is why both SVR and NN regress on `log1p(count)` rather than raw counts.

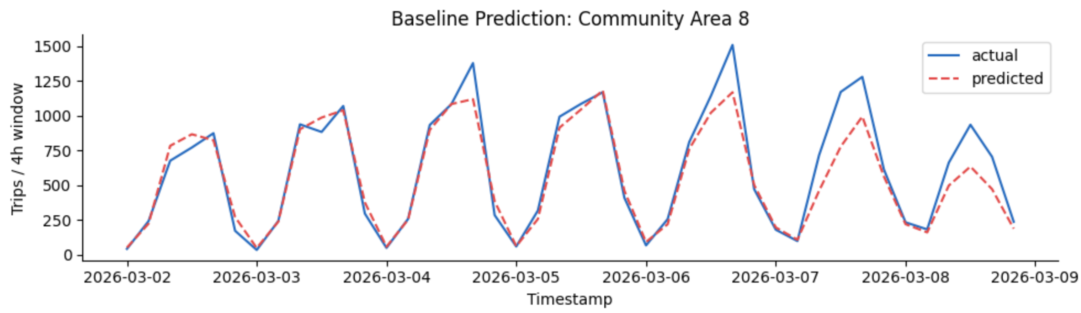

Figure 4: Profile baseline predictions vs. actual demand across an evaluation week, highest-demand community area (community area, 4-hour resolution). The baseline tracks the daily and weekly cycle closely.

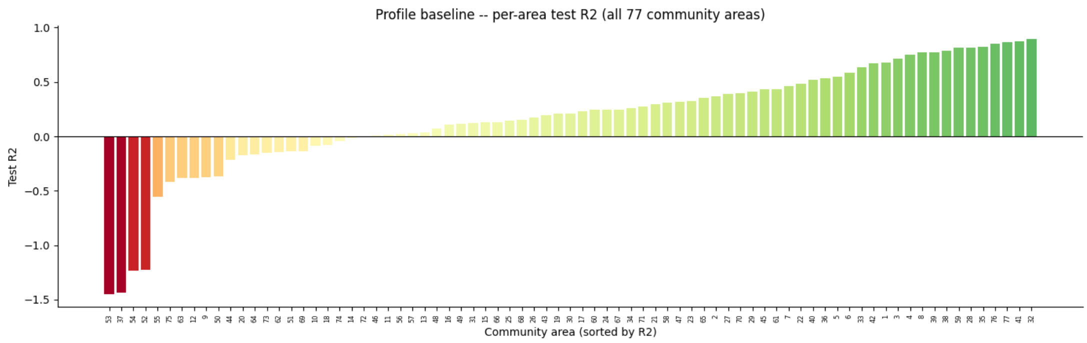

Figure 5: Profile baseline test R^2 per community area, 4-hour resolution. The pooled R^2 (0.933) is dominated by the highest-volume areas; the per-area distribution is much wider, with 21 of 77 areas negative.

Table 8: Profile baseline test scores by resolution.

Resolution	Units	Pooled R^2	Weighted RMSE	Median per-unit R^2	Units with $R^2 < 0$
Community area, 4-hour	77	0.933	139.02	0.208	21 of 77
Community area, 1-hour	77	0.918	39.46	0.028	36 of 77
Census tract, 4-hour	179	0.896	100.99	0.195	56 of 179

8.5 Predictive Analytics: Community Area, 4-Hour Resolution

Table 9: SVR test scores, community area, 4-hour resolution. Per-area models cover only the 20 areas above the demand threshold.

Model	Units	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	77	0.000	0.000	139.02	0.933

Model	Units	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
LinearSVR, pooled	77	-0.827	-0.017	254.94	0.778
LinearSVR, demand-weighted	77	-0.675	-0.052	231.91	0.814
LinearSVR, demand-tiered	77	-0.732	0.000	243.60	0.800
RBF, tuned (10k subsample)	77	-0.232	-0.027	176.93	0.892
Polynomial, tuned (10k subsample)	77	-0.221	-0.049	171.71	0.896
LinearSVR, per-area	20	-0.112	-0.091	156.93	0.905
LinearSVR, per-area, tuned	20	-0.125	-0.116	161.10	0.901
RBF, per-area	20	-0.135	-0.105	163.37	0.897
RBF, per-area, tuned	20	**+0.012**	-0.002	143.80	0.921

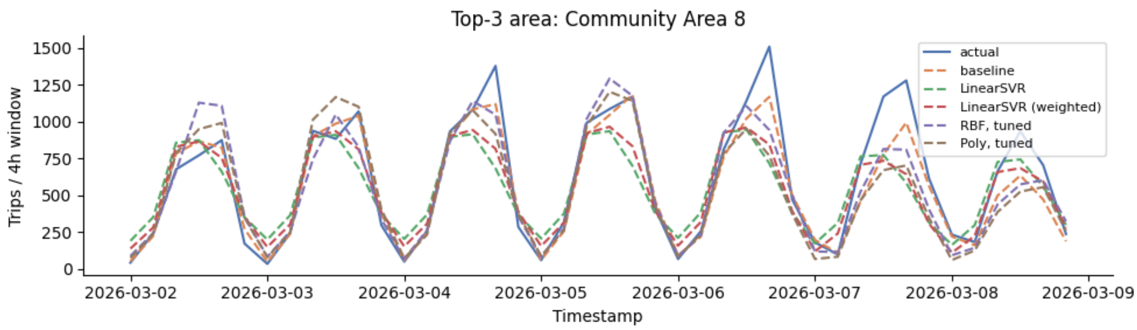

Figure 6: Actual, baseline and SVR-variant demand across the March evaluation week, Community Area 8, community area 4-hour resolution.

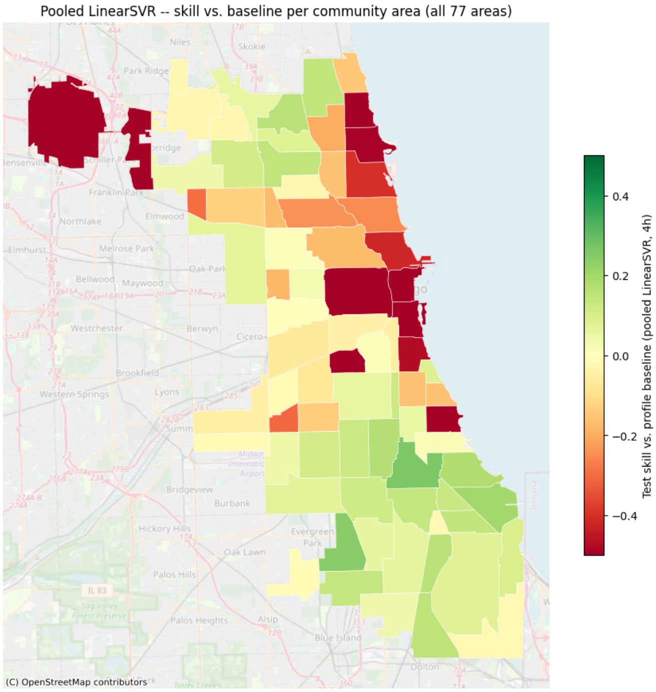

Figure 7: Pooled `LinearSVR` per-area skill vs. the profile baseline, all 77 community areas, 4-hour resolution.

Table 10: Neural network test scores, community area, 4-hour resolution.

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2	Fit time
Profile baseline	0.000	0.000	139.02	0.933	1s
Shallow NN (untuned)	-0.028	+0.097	144.96	0.925	324s
Deep NN (tuned)	-0.027	+0.071	145.93	0.927	2,333s

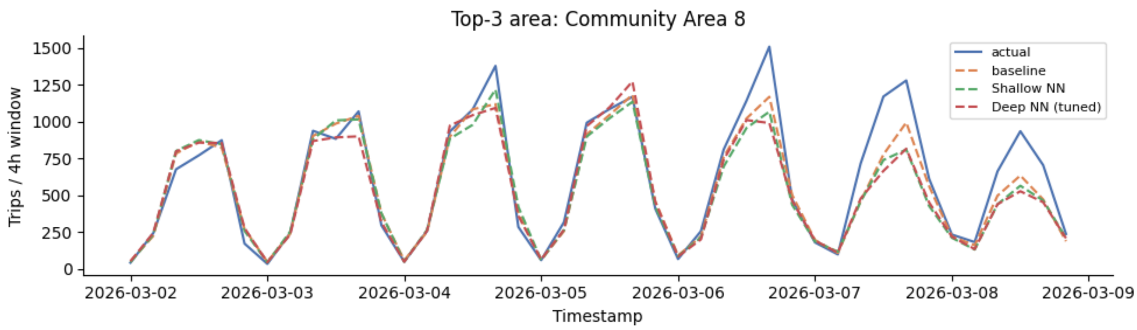

Figure 8: Actual, baseline and neural-network demand across the March evaluation week, Community Area 8 (highest demand), community area 4-hour resolution.

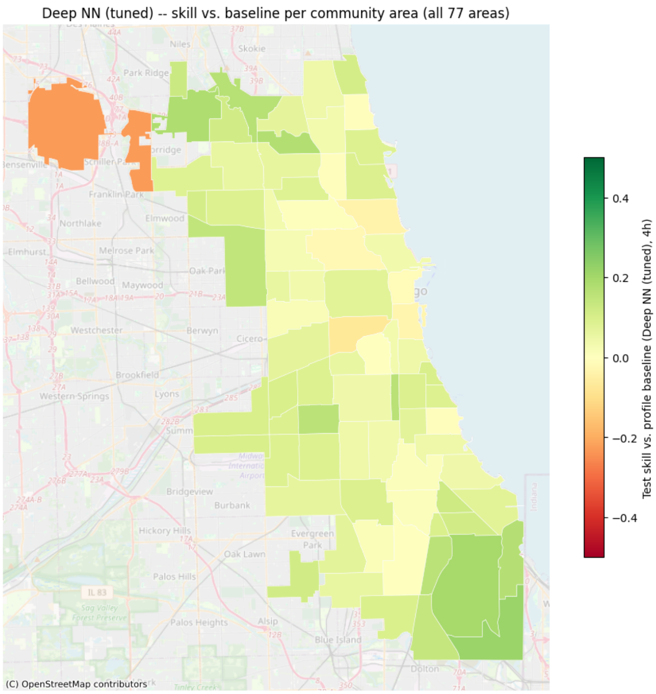

Figure 9: Tuned deep NN per-area skill vs. the profile baseline, all 77 community areas, 4-hour resolution. O'Hare (upper left) is the one clearly negative area.

8.6 Predictive Analytics: Community Area, 1-Hour Resolution

Table 11: Test scores for both families, community area, 1-hour resolution.

Model	Dem.-weighted skill	Median skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	0.000	39.46	0.918
SVR – LinearSVR, pooled	-1.016	+0.022	76.96	0.687
SVR – RBF, tuned	-0.331	+0.015	54.53	0.835
SVR – Polynomial, tuned	-0.361	0.000	56.14	0.793
NN – Shallow (untuned)	-0.048	+0.118	42.28	0.903
NN – Deep (tuned)	**+0.062**	+0.090	36.26	0.928

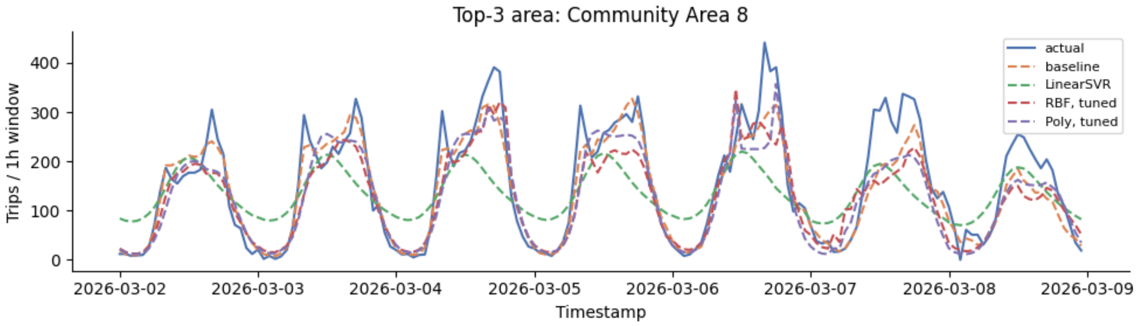

Figure 10: SVR March-week demand, Community Area 8, 1-hour resolution.

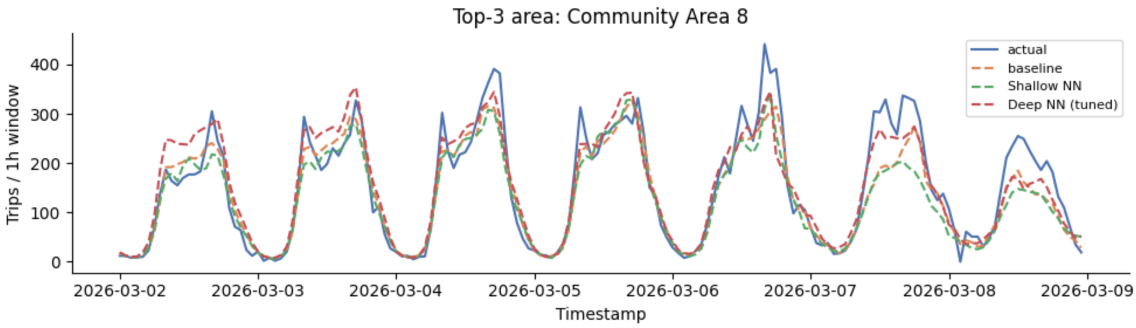

Figure 11: Neural-network March-week demand, Community Area 8, 1-hour resolution.

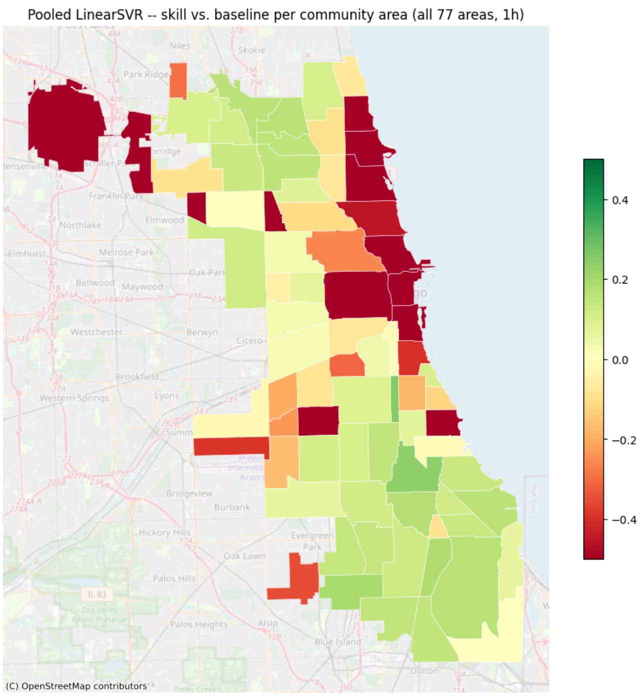

Figure 12: Pooled `LinearSVR` per-area skill vs. the profile baseline, all 77 community areas, 1-hour resolution.

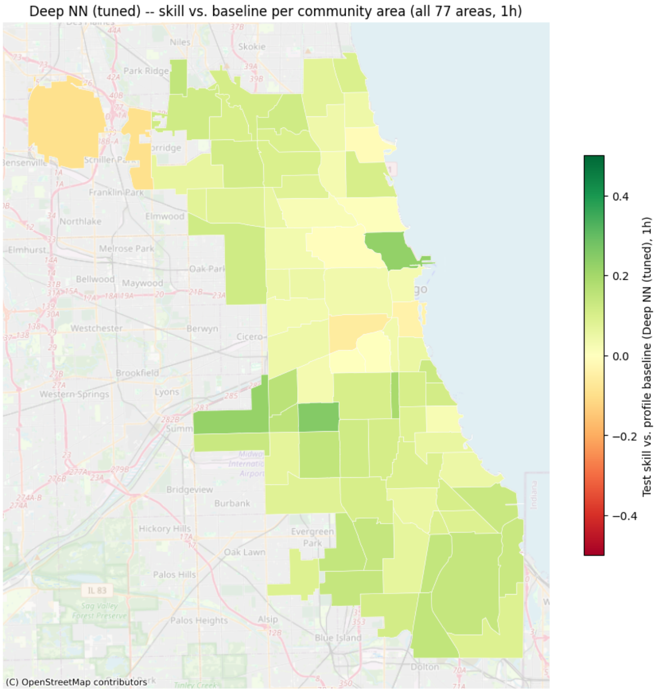

Figure 13: Tuned deep NN per-area skill vs. the profile baseline, all 77 community areas, 1-hour resolution.

8.7 Predictive Analytics: Census Tract, 4-Hour Resolution

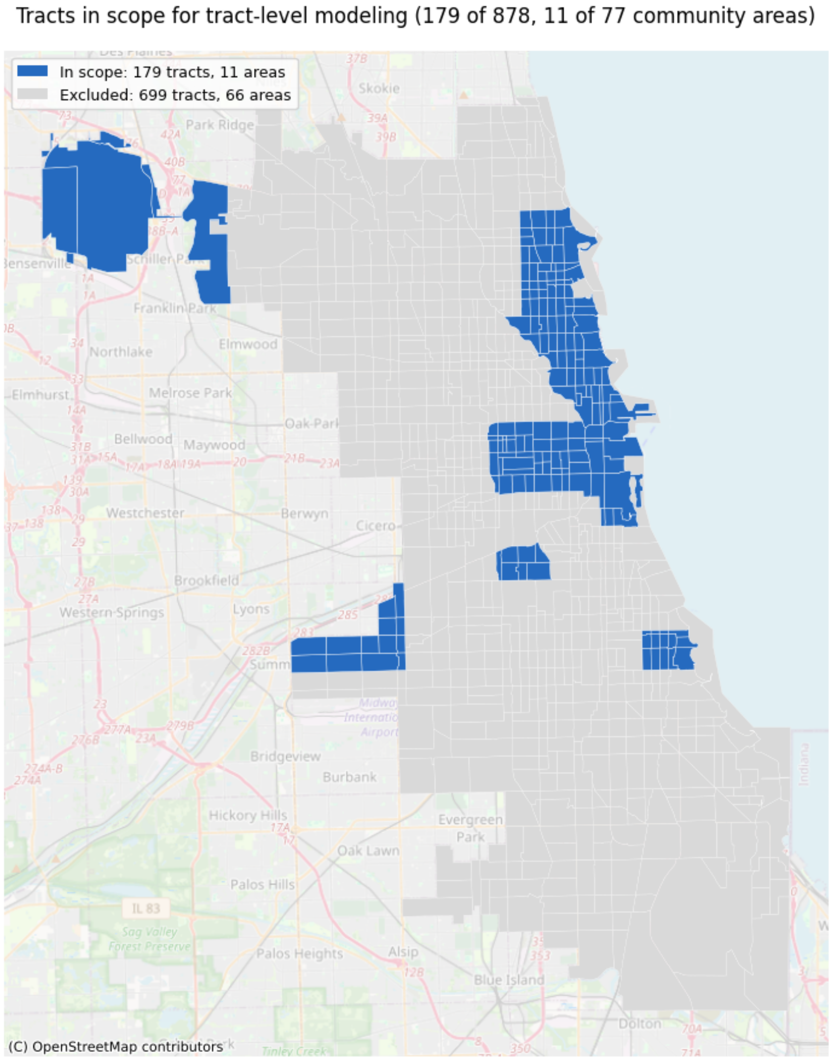

Figure 14: Scope map: the 179 of 878 census tracts with enough tract-precision data to support a tract-level model, across 11 community areas.

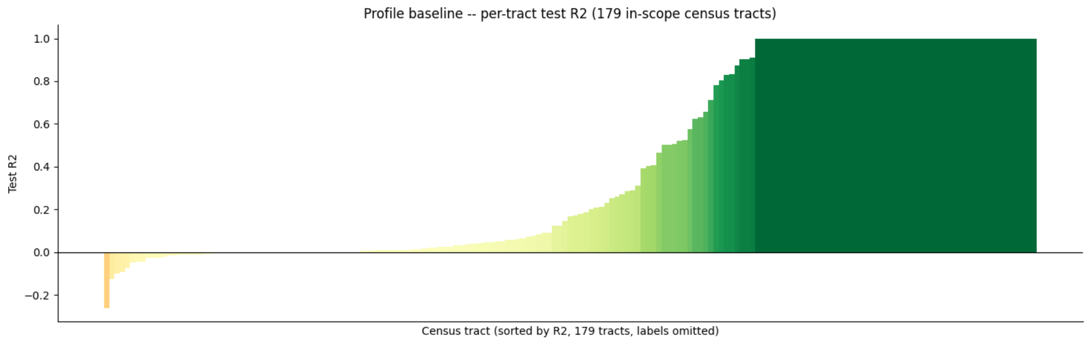

Figure 15: Profile baseline test R^2 per census tract, 4-hour resolution. Wider spread than the community-area version (Figure 5), consistent with more units competing for the same total demand.

Table 12: Neural network test scores, census tract, 4-hour resolution (179 in-scope tracts).

Model	Dem.-weighted skill	Weighted RMSE	Pooled R^2
Profile baseline	0.000	100.99	0.896
Shallow NN (untuned)	-0.098	113.77	0.868
Deep NN (reused 4h config)	-0.340	138.71	0.821

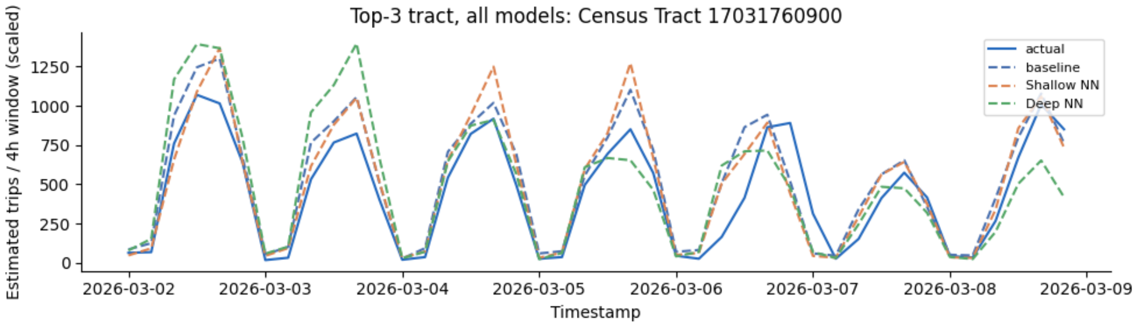

Figure 16: Actual, baseline and neural-network demand across the March evaluation week, top-demand census tract (17031760900), 4-hour resolution (estimated trips after inverse-retention scaling).

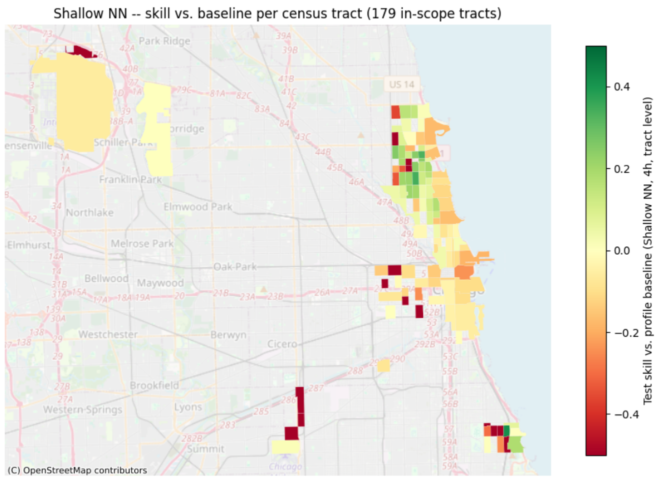

Figure 17: Tuned deep NN per-tract skill vs. the profile baseline, 179 in-scope census tracts, 4-hour resolution.

8.8 Reinforcement Learning: Environment Assumptions

Table 13: Environment assumptions for the charging task

Quantity	Value	Meaning
Battery capacity C	{40, 60, 77} kWh	uniform per episode – compact, mid-size, and large vehicle classes

Quantity	Value	Meaning
Initial SOC	$\mathcal{U}$(20%, 60%) of C	varies daily with shift length and route profile
Charging window	8 × 15 min	14:00-16:00
Charging powers	{0, 5, 11, 22} kW	zero / low / medium / high (later refined, see Markov Decision Process)
Energy demand d	$\mathcal{N}(30, 5^2)$ kWh, clipped at C	trip demand depends on the route, not on vehicle size – revealed only at departure
Penalty	100 €/kWh shortage	shortfall penalty

8.9 Reinforcement Learning: Results

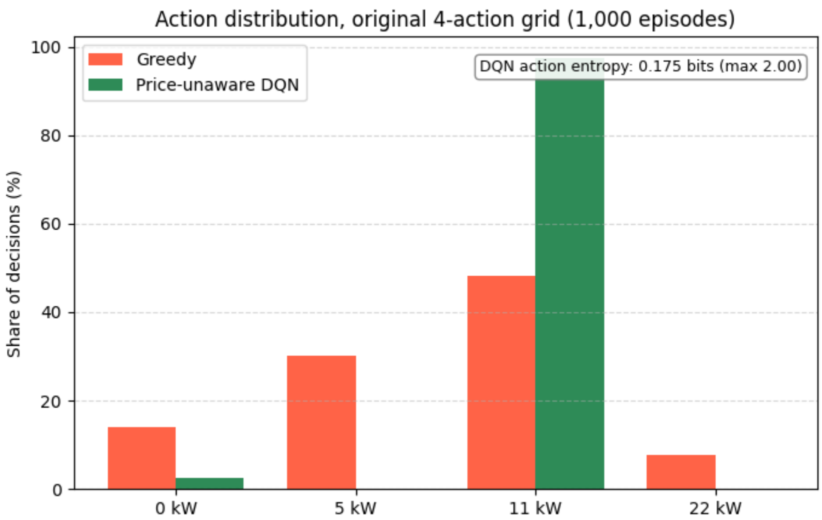

Figure 18: Action distribution on the original 4-action grid, 1,000 episodes (price-unaware DQN retrained with the notebook's original hyperparameters and seed for this comparison). Greedy spreads its decisions across all four power levels, while the DQN collapses onto 11 kW in 97% of them – the degenerate habit that the refined grid in Table 14 corrects.

Table 14: Effect of action-grid resolution in the stylized environment, 1,000 episodes

Grid	Policy	Success rate	Mean reward
Original (4 actions)	Greedy	98.9%	-2.581
Original (4 actions)	Price-unaware DQN	95.5%	-17.201
Refined (7 actions)	Greedy	98.9%	-2.058
Refined (7 actions)	Price-unaware DQN	98.3%	-1.855

Table 15: Policy comparison under EPEX prices, refined action grid

Policy	Success rate	Mean cost (€/day)	Mean reward
Greedy	98.9%	1.7223	-3.7775
Price-unaware DQN (stylized-trained)	98.3%	2.0424	-3.8872
EPEX DQN	75.2%	0.9931	-88.0377
Oracle (40 episodes)	100%	0.87	–

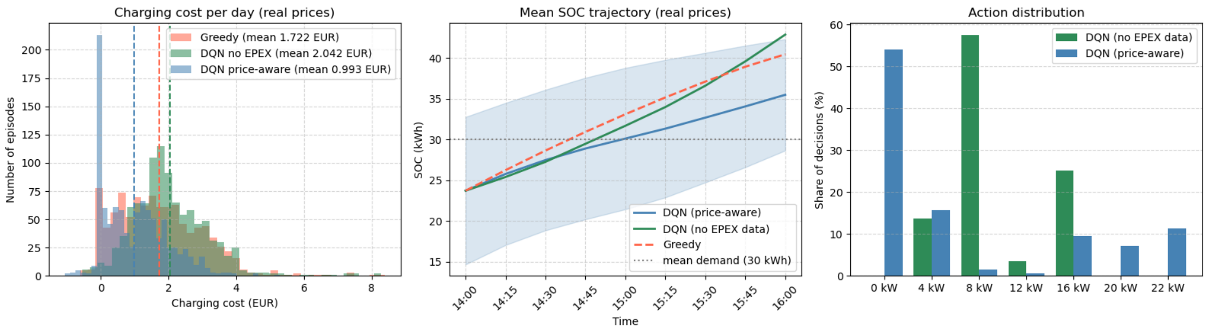

Figure 19: Charging-cost distribution, mean SOC trajectory, and action mix under EPEX prices, refined grid, 1,000 paired episodes. The EPEX DQN's cost mass sits near zero and its SOC trajectory runs lowest of the three policies, consistent with its lower success rate in Table 15; the price-unaware DQN concentrates on 8 kW and 16 kW, while the EPEX DQN uses more of the grid, including 0 kW.

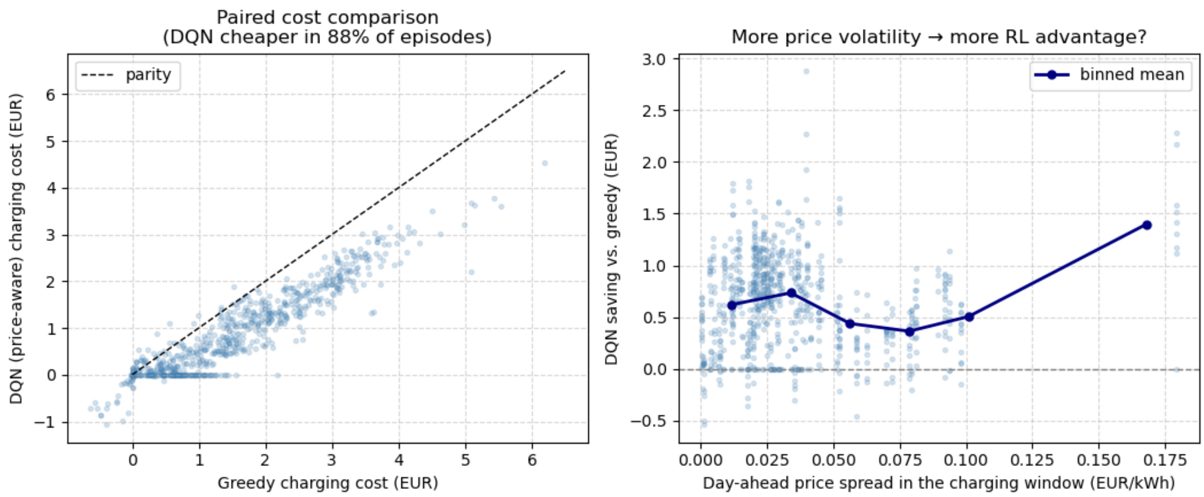

Figure 20: Left: paired charging cost, EPEX DQN vs. greedy, on the 752 episodes both cover – points below the parity line are episodes where the EPEX DQN is cheaper. Right: the EPEX DQN's saving over greedy grows with the day-ahead price spread within the charging window, indicating that the advantage comes from price timing rather than a fixed offset.

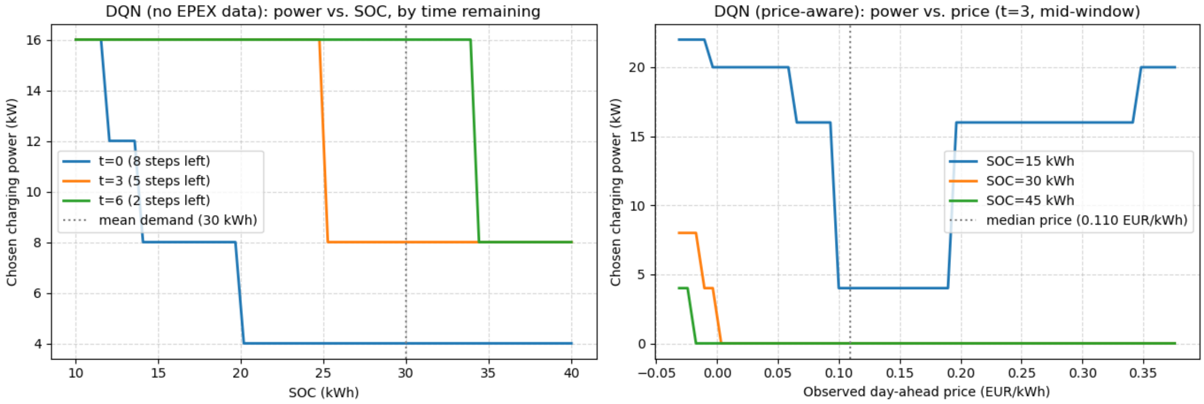

Figure 21: Learned charging policies as a function of state, refined grid. Left: the price-unaware DQN's chosen power falls as SOC rises and as fewer steps remain, independent of price. Right: the EPEX DQN's chosen power falls as price rises above the median (dotted line) and shifts up as SOC drops from 45 to 15 kWh, so urgency dominates price once charge is low.

References

Fraunhofer Institute for Solar Energy Systems ISE. (2026). *Energy-charts: Day-ahead electricity prices* [Data sourced from ENTSO-E / EPEX SPOT, DE-LU bidding zone, license CC BY 4.0]. Retrieved July 20, 2026, from https://www.energy-charts.info